The following problem involves the number a_2, which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
      k=k+1
      sum = sum + (1/2)^(k) - 72^(-1000000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided ${0.4, 0.6}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?
Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 100 seconds

Yes.

The code defines a_2 as the limit of the sequence output by `Compute_a_2(n)` as $n \to \infty$. Analyzing the loops shows that

$$a_2 = 1 + 2\varepsilon,$$

where $\varepsilon = 72^{-B} - \frac{72^{-3B}}{1-72^{-B}}$ and $B = 10^{24}$. Because B is huge, $\varepsilon > 0$ is unimaginably small, so $a_2 > 1$ but extremely close to 1.

The optimization problem is to minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ and $x_1, x_2 \geq 0$. Since $a_2 > 1$, the unique minimizer is $(x_1^*, x_2^*) = (0,\ 1/a_2)$. Because $a_2 \approx 1$, $1/a_2 \approx 1$.

The infinity-norm distance from the proposed point $(0.4, 0.6)$ to the true minimizer is

$$\max\big(|0.4 - 0|,\ |0.6 - 1/a_2|\big) = \max\big(0.4,\ 1/a_2 - 0.6\big) = 0.4,$$

since $1/a_2 < 1$ and therefore $1/a_2 - 0.6 < 0.4$.
This distance 0.4 is indeed at most 0.45.